BREAKING THE CARBON MARKET LOGJAM:

1. Tokenization

2. Liquidity Pools

3. Decentralized Price Discovery

How Tokenization, Liquidity Pools, and Decentralized Price Discovery Can Transform South Africa's Carbon Trading System

A Strategic Transformation Framework for 2024

Breaking the Carbon Market Logjam:

How Tokenization, Liquidity Pools, and Decentralized Price Discovery Can Transform South Africa's Carbon Trading System

A Discussion Paper Exploration into Market Infrastructure Gaps and Umbane's Solution

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[please note: This is a draft version]

Executive Summary

The South African carbon credit market faces a critical infrastructure paradox: robust regulatory frameworks exist -- Carbon Tax Act, Carbon Offset Administration System (COAS) registry, JSE Ventures Carbon Market, zaREC registry and Renewable Energy Certificate Market Participants' Association of South Africa (RECSA) -- yet the market suffers from severe liquidity constraints, opaque pricing, and monopolistic intermediation. This paper analyzes the root causes of market friction and proposes a tokenization-based solution leveraging blockchain technology to create automated market makers (AMMs), transparent price discovery, and direct market access for small-scale domestic energy producers.

Key Findings:

1. Centralization Bottleneck: Single-vendor dependency (Xpansiv/Evident) on JSE creates market friction and limits API/SDK access

2. Registration Barrier: Domestic solar producers must register devices either directly via I-REC administered by UK-based Green Certificate Company (GCC) or via the zaREC registry, before accessing carbon markets and most onchain solutions like Enerlink, exist external to the country. There is apparently a localised project startup in George engaged in blockchain activity.

3. Price Opacity: Carbon credit prices vary wildly with no transparent, real-time price feed

4. Liquidity Desert: JSE is not a market maker; it provides infrastructure but no liquidity guarantees

5. Exclusion of Small Producers: High registration costs R25 000 ex Vat to register a device with zaREC and minimum transaction fee ~R4,600 (ex

VAT) and associated complexity immediately exclude household solar prosumers and township microgrids from carbon markets.

6. Techno-Feudalism: Currently the only rewards for domestic energy production are the feed-in tariffs set by City Power, with no possibility of collective bargaining for housing schemes and microgrid projects.

†

Umbane's Proposition: Create a parallel tokenized carbon credit system “UMB” (UMB wrapped mJ → aC tokens) that:

- Aggregates I-REC/D-REC and or zaREC registration for domestic-

scale producers

- Provides automated market making via DeFi liquidity pools
- Enables transparent, real-time price discovery
- Offers API/SDK access for frictionless integration
- Bridges to traditional carbon markets (JSE, COAS) via verifiable oracles
- Interfaces with I-REC/D-REC and/or zaREC as an aggregator of renewable energy.

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1. The Current State of South African Carbon Markets

1.1 Regulatory Framework (2026)

South Africa's carbon market operates under multiple overlapping frameworks:

Compliance Market (Mandatory):

- Carbon Tax Act (2019): R190/ton CO2 as of 2026, increasing annually
- Carbon Offset Administration System (COAS): National registry for tax-eligible offsets
- Eligible Standards: Verra (VCS), Gold Standard, CDM
- Geographic Requirement: Only SA-based projects can generate tax-eligible credits
- Offset Limits (Phase 2, 2026):
  - 10% for fugitive/process emissions
  - 15% for combustion emissions
- Demand Estimate: 10-14 *MtCO2e* annually

Voluntary Market (Optional):

- JSE Ventures Carbon Market (launched 2023, powered by Xpansiv)
- Credible Carbon Registry: SA-focused, small-scale projects
- I-REC Registry (Evident.app): Renewable Energy Certificates
- International Standards: Integration with Verra, Gold Standard, ACR, CAR
- zaRECs (Pty) Ltd: administers the South African voluntary REC market along the lines of the EECS specifications on behalf of members of the

voluntary Renewable Energy Certificate South Africa market participant's association (RECSA).

- RECSA: a non-profit body that supports the issuance, trading and redemption of Renewable Energy Certificates (RECs) in South Africa.

1.2 Market Statistics (2026)

Supply Side:

- 14.2 million carbon credits issued by SA projects (Verra + Gold Standard) as of June 2025 [1]
- 10.4 million retired (2019-2025)
- Annual issuance doubled in 2024 vs 2023
- Concentration: Single clean cookstove project issued 67% of 2024 supply (2.7M credits)
- Sectoral Breakdown:
  - Energy demand (cookstoves, efficiency): 42%
  - Forestry/AFOLU: 28% (estimated from sector analysis)
  - Waste: 18% (estimated from sector analysis)
  - Transport: 12% (estimated from sector analysis)

Demand Side:

- Compliance demand: ~10-12 MtCO2e/year (carbon tax offsets)
- Voluntary demand: Corporate net-zero commitments (Nedbank, Aurecon, TMNP)
- Gap: Supply significantly lags demand

Pricing (JSE Ventures Carbon Market - spot prices):

- Generic VCU (Verra): R120-R180/ton CO2 ($7-$10 USD)
- Gold Standard cookstove: R200-R300/ton CO2 ($11-$17 USD)
- I-REC (South Africa wind/solar): R8-R15/MWh ($0.44-$0.83 USD)
- Volatility: ±40% price swings within 90 days (no transparent benchmark)

1.3 The JSE's Role (And Its Limitations)

Following a meeting with Chris Sturgess (JSE Carbon Trading Desk), it appears the JSE is not a market maker:

What JSE Provides:

- Trading infrastructure, white-label Xpansiv CBL platform. CBL is "*world's largest and most active spot marketplace for environmental commodities".*
- Regulated marketplace (oversight, reporting)
- Connection to international registries (Verra, Evident, COAS)
- Post-trade settlement automation

What JSE Does NOT Provide:

- Liquidity guarantees (no standing buy/sell orders)
- Price stabilization (no inventory to buffer volatility)
- API/SDK for direct integration (reliance on Xpansiv proprietary systems)
- Market-making services (buyers and sellers must find each other)

Result: Thin markets, wide bid-ask spreads, episodic trading volumes.

1.4 Key Pain Points (Based on JSE Carbon Conversation)

1. Carbon credits "don't appear to be easily tradeable"
2. Prices "all over the place" - No transparent benchmark, regional variation
3. Single-vendor lock-in: Xpansiv/Evident monopoly creates dependency
4. No public API/SDK: Programmatic access limited to Xpansiv clients
5. High entry barriers: I-REC /zaREC device registration too complex/expensive for household prosumers and small scale projects.

2. Anatomy of Market Friction

2.1 The Xpansiv Bottleneck

Xpansiv Infrastructure (powers JSE Ventures Carbon Market):

- CBL (Carbon + Biodiversity Ledger): Global spot exchange
- Xpansiv Connect: Multi-registry portfolio management
- Evolution Markets: Brokerage arm

Advantages:

- Deep international liquidity (330M+ credits traded globally)
- Integration with major registries
- Standardized contracts (GEO, N-GEO, C-GEO)

Disadvantages (South African Context):

- Proprietary Platform: No open API for third-party developers
- Data Access: Market data via paid subscriptions (Xpansiv Data™)
- Membership Barriers: KYC/AML requirements, minimum capital

- Liquidity Asymmetry: Global liquidity doesn't guarantee local SA liquidity
- Single Point of Failure: Xpansiv downtime = JSE market downtime

2.2 The Lack of Market Makers

Traditional Market Maker Functions:

1. Continuous Quotes: Always willing to buy/sell at stated prices
2. Inventory Management: Hold inventory to buffer supply/demand shocks
3. Price Stabilization: Dampen volatility through strategic trading
4. Liquidity Provision: Ensure trades execute quickly with tight spreads

Why No Market Makers in SA Carbon Market?:

- Regulatory Uncertainty: Carbon assets not clearly defined under Financial Sector Conduct Authority (FSCA) rules
- Inventory Risk: Carbon credit quality/vintage variations make inventory risky
- Capital Requirements: Market making requires significant capital reserves
- Low Volumes: Insufficient trading volume to justify MM operations
- Technology Gap: Lack of real-time pricing infrastructure

Result: Carbon credits trade like illiquid OTC bonds, not liquid commodities.

2.3 Price Discovery Failure

What is Price Discovery?

The process by which markets determine the fair value of an asset through:

1. Information aggregation (buyers/sellers reveal preferences)
2. Order flow (continuous trading reveals supply/demand)
3. Transparent order books (all participants see bids/offers)

Why SA Carbon Market Fails at Price Discovery

Requirement	JSE Ventures Status	Impact
Continuous Trading	Episodic (days between trades)	Stale prices
Transparent Order Book	Limited visibility (Xpansiv clients only)	Information asymmetry
Sufficient Volume	Low (10-50 trades/month estimated)	Thin market
Homogeneous Assets	Highly differentiated (project, vintage, standard)	No fungibility
Low Transaction Costs	High (registry fees + platform fees)	Discourages trading
Real-Time Data Feeds	No public API	Cannot build price indices

4. Arbitrage (price differences are exploited, converging prices)

Observation from seeing JSE Carbon Desk spreadsheet: "Prices all over the place" – This is textbook price discovery failure.

2.4 Comparison: Traditional Finance vs Current Carbon Market

Market Infrastructure Comparison: JSE Equities vs. SA Carbon Market

Feature	Stock Market (JSE Equities)	SA Carbon Market (JSE Ventures)
Market Makers	Multiple (banks, prop firms)	None
Liquidity	Deep (millions of shares/day)	Shallow (sporadic trades)
Price Updates	Real-time (every second)	Delayed (days between prices)
API Access	Yes (JSE Market Data API)	No (Xpansiv proprietary)
Spreads	Tight (<0.5%)	Wide (10-30%)
Settlement	T+3	T+0 (but rare trades)
Fragmentation	Centralized (JSE)	Fragmented (multiple registries)

Conclusion: SA carbon market is pre-electronic trading era in terms of infrastructure maturity.

3. The I-REC / zaREC Paradox

3.1 What is I-REC?

I-REC (International Renewable Energy Certificate):

- Tracks 1 MWh of renewable energy generation
- Used to prove "green" electricity procurement
- Required for corporate sustainability reporting (CDP, RE100, SBTi)
- *Different from carbon credits:* I-RECs prove RE usage; carbon credits prove emission reductions

I-REC Registry: Evident.app (formerly APX)

- Global registry system
- Issues I-RECs for registered devices/projects
- Tracks transfer, retirement of certificates

3.2 The South African I-REC System

Central Issuer for South Africa: GCC (The Green Certificate Company)

- *Location*: Sheffield, UK (400 Springvale Road, S10 1LP)
- *Role:* Sole authorized issuer of I-RECs in South Africa
- Process:
  1. Device owner registers with GCC
  2. GCC verifies device meets criteria
  3. GCC issues I-RECs based on metered production
  4. I-RECs listed on Evident registry

Devices Registered in South Africa (per Evident Device Register):

- 19+ devices in Cape Town metro alone

- Primarily large-scale: shopping malls, factories, solar farms (1-10 MW), wind farms

- Zero household/domestic solar installations

- Zero collective housing schemes

- Zero township microgrid projects

3.3 The zaREC System

- 12 Active Market Makers, including:

- Borena Energy
- Carbon Tradex
- Energy Exchange Southern Africa (Pty) Ltd.
- Fuel Switch
- Green Asset Exchange
- Independent Energy Pool (IEP) GmbH
- POWERX Energy

- 179 Customers (Organisations Purchasing RECS), including:

Private Companies

- ABSA Group Limited
- AdWiser
- African Sun Energy (Pty) Ltd
- Aldus Renewable Energy
- Autoneum Feltex (PTY) Ltd.

- Delheim Wines
- Enertrag South Africa (Pty) Ltd
- Full Circle Contact Centre Services (Pty) Ltd t/a Capita South Africa
- G7 Renewable Energies
- Ibert (Pty) Ltd.
- Ideas Wise and Wonderful
- IF Engineering
- JCDecaux Sub-Saharan Africa
- JT International South Africa (Pty) Ltd.
- MA Automotive Tool & Die (Pty) Ltd
- Marketecture CC t/a Carbon Tradex
- Maxion Wheels South Africa (Pty) Ltd
- Mondelez (Pty) Ltd.
- Nedbank Limited
- Old Mutual Emerging Markets (Pty) Ltd.
- Old Mutual Life Assurance Company of South Africa
- Promethium Carbon
- RCL Food Sugar and Milling (Pty) Ltd.
- Red Rocket South Africa (Pty) Ltd
- Sibanye Energy
- VdW & Co

- Vineyard Hotel and Spa
- Volkswagen of South Africa (Pty) Ltd

Events

- Rocking the Daisies Festival

Development Agencies

- GIZ

Societies & Associations

- Green Building Council of SA

- 3 Affiliates including:

- VdW & Co

3.3 Why Household Prosumers Are Excluded

Registration Barriers:

1. Cost: estimated R15,000-R50,000 ($830-$2,770) IREC registration fee per device, R25 000 (ex VAT) zaREC registration fee per device.

2. Minimum Transaction Fees, zaREC have a minimum transaction fee ~R4,600 (ex VAT)

2. Documentation: Metering verification, grid connection proof, ownership documentation, requiring consultants and third party administration

3. Minimum Size: Effectively 100 kW+ (household solar typically 3-10 kW)

4. Annual Fees: R5,000-R10,000/year registry maintenance

5. Complexity: Requires consultant/agent familiar with I-REC/ zaREC system

Economic Reality:

- Household 5 kW solar system produces ~8,000 kWh/year
- 8,000 I-RECs × R10/REC = R80,000/year gross value
- Registration + annual fees = R25,000
- Net: R55,000/year (if one can even find buyer for small lots)
- ROI: Not worth hassle for individual homeowner
- not viable for microgrid users in Khayalitsha.

JSE Carbon Desk Perspective:

"Where are the I-REC/zaRECs for your devices" – This may be characterised as a ‘gatekeeper mentality’. Without an I-REC or zaREC, you cannot participate in formal carbon/REC markets via JSE and other exchanges.

3.5 The Umbane Alternative

What Umbane Proposes:

Instead of forcing 10,000 Cape Town households to each register with GCC Sheffield or zaREC create a parallel tokenized system that can bridge or wrap I-REC/D-REC/zaREC

1. Device Registration: One-time onboarding via Umbane (R500)
2. Metering: Low-cost IoT devices (R1,200-R2,100) with tamper-proof signatures
3. Tokenization: Production → mJ tokens (instantly, on-chain)

4. Carbon Credits: mJ → aC tokens (verified via Chainlink oracle + DAO verification)

5. Liquidity: aC tokens tradeable on DeFi liquidity pools (24/7, no intermediaries)

6. Bridge: aC tokens can be "wrapped" and submitted to COAS/JSE (if desired)

Key Difference: Umbane dis-intermediates GCC & zaREC and creates bottom-up carbon credits from household solar production, rather than top-down REC issuance.

4. Price Discovery Mechanisms: TradFi vs DeFi

4.1 Traditional Finance (TradFi) Price Discovery

Order Book Model (JSE Equities, Xpansiv CBL):

BUY ORDERS (Bids)		
Price	Quantity	Trader
R175	500	Buyer A
R174	1,000	Buyer B
R173	2,500	Buyer C

```
│       SELL ORDERS (Asks)          │
├──────────┬──────────┬──────────────┤
│ Price    │ Quantity │ Trader       │
├──────────┼──────────┼──────────────┤
│ R180     │ 1,500    │ Seller X     │
│ R181     │ 2,000    │ Seller Y     │
│ R185     │ 5,000    │ Seller Z     │
└──────────┴──────────┴──────────────┘
```

Spread: R180 - R175 = R5 (2.8% spread)

```

**Requirements for Effective Price Discovery:**

1. Many Participants: Hundreds of active traders
2. Continuous Orders: Orders constantly added/removed
3. Market Makers: Firms that narrow the spread (e.g., buy at R174, sell at R181)
4. Transparent Book: All participants see all orders
5. Fast Matching: Sub-second order execution

**SA Carbon Market Reality:**

- Few participants (~50-100 active traders)
- Sporadic orders (days between trades)
- Few market makers
- Limited book transparency (Xpansiv clients only)
- Slow matching (manual RFQs common)
```

Result: Wide spreads (10-30%), stale prices, illiquid market.

4.2 DeFi Price Discovery via Automated Market Makers (AMMs)

Liquidity Pool Model (Uniswap, SushiSwap, Curve):

Instead of order books, AMMs use liquidity pools + bonding curves.

Core Concept:

```
┌──────────────────────────────────────────┐
│        LIQUIDITY POOL                    │
│                                          │
│  10,000 aC tokens   ←→   R1,750,000      │
│  (Carbon Credits)       (ZAR stable)     │
│                                          │
│  Ratio: 1 aC = R175                      │
└──────────────────────────────────────────┘
```

Trading Mechanism:

- Buy aC: User deposits ZAR → pool adjusts ratio → user receives aC
- Sell aC: User deposits aC → pool adjusts ratio → user receives ZAR
- Price: Determined by ratio in pool ($x \times y = k$ constant product formula)

Example Trade:

BEFORE TRADE:

Pool: 10,000 aC × R1,750,000 = k = 17,500,000,000

USER BUYS 100 aC:

User deposits: R17,675 (calculated to maintain k)

New pool: 9,900 aC × R1,767,675 = k = 17,499,986,500 ≈ constant

New price: R1,767,675 / 9,900 = R178.55/aC

Price Impact: R175 → R178.55 = 2% slippage

Key Advantages Over Order Books:

1. Always Available: Pool never "runs out" of liquidity
2. Algorithmic Pricing: No need for market makers (math does it)
3. Transparent: Formula is public, anyone can calculate price
4. Permissionless: Anyone can trade or add liquidity
5. Composable: Pools can be integrated into other DeFi apps

4.3 AMMs Applied to Carbon Credits

Successful Examples:

KlimaDAO / Toucan Protocol (Polygon):

- 25M carbon credits bridged to blockchain (TCO2 tokens)
- 17.5M locked in KlimaDAO treasury
- Liquidity pools: BCT/USDC, MCO2/USDC on SushiSwap
- $10M+ daily volume at peak (2021-2022)

- Price Discovery: Real-time, transparent, API-accessible

Carbonmark (Polygon):

- Marketplace + API for tokenized carbon credits
- AMM-based pricing for fractional credits
- Integration with Verra registry
- Automated retirement via smart contracts

C3 Protocol (Polygon):

- Permissionless carbon bridge
- ERC-20 pools backed by NFTs (individual credits)
- Curve pools for deep liquidity on similar vintages

Market Infrastructure: TradFi (JSE Ventures) vs. DeFi (AMM Pools)

Problem	TradFi (JSE Ventures)	DeFi (AMM Pools)
Liquidity	Depends on market makers	Algorithmic, always available
Price Updates	Sporadic (manual trades)	Continuous (every block ~2 sec)
Spreads	Wide (10-30%)	Tight (0.3-1% with deep pools)
Accessibility	Xpansiv membership required	Anyone with wallet
API Access	Proprietary (Xpansiv Data™)	Open (read blockchain)
Market Making	Requires specialized firms	Anyone can provide liquidity
Transparency	Limited order book visibility	Full on-chain visibility
Settlement	T+0 but manual	Instant (atomic swaps)
Operating Hours	Business hours	24/7/365

4.4 Why AMMs Solve the Carbon Market Problem

JSE Carbon Desk Concern Addressed:

- "Prices all over the place" → AMMs provide algorithmic consensus pricing
- "Not easily tradeable" → AMMs enable instant swaps via wallet
- No API/SDK → AMMs are native APIs (smart contract calls)

5. Liquidity Pools and Automated Market Makers (Deep Dive)

5.1 How Liquidity Pools Work

Basic Liquidity Pool (Constant Product AMM):

Formula: `x × y = k` (Uniswap V2 model)

Where:
- `x` = quantity of Token A (e.g., aC carbon credits)
- `y` = quantity of Token B (e.g., USDC stablecoin)
- `k` = constant (doesn't change during trades)

Example: aC/USDC Pool

Initial Liquidity:
x = 100,000 aC tokens
y = $175,000 USDC
k = 17,500,000,000

Price: y/x = $175,000 / 100,000 = $1.75 per aC

Trade: User buys 1,000 aC

1. User deposits USDC (amount calculated to maintain k)
2. Pool gives user 1,000 aC
3. New balances:
   x = 99,000 aC
   y = 176,768 USDC (calculated: k / 99,000)
   New price: $176,768 / 99,000 = $1.79 per aC

Slippage: ($1.79 - $1.75) / $1.75 = 2.3%

Key Insight: Large trades cause price impact (slippage). The larger the pool, the lower the slippage.

5.2 Liquidity Provision (LP) Economics

Who Provides Liquidity?

Anyone can be a liquidity provider (LP) by depositing equal value of both tokens into the pool.

Example:

- Alice deposits: 10,000 aC + R17,500 ZAR
- Pool issues: LP tokens representing Alice's share
- Alice earns: Trading fees (e.g., 0.3% per swap)

Returns:

Pool Stats:

Daily Volume: $50,000

Fee: 0.3%

Daily Fees: $150

Alice's Share: 10% of pool

Alice's Daily Earnings: $15

Annual Yield: ($15 × 365) / $17,500 = 31.3% APY

Risks:

1. Impermanent Loss: If aC price moves significantly, LP loses vs just holding

2. Smart Contract Risk: Bugs in AMM code could drain pool

3. Token Risk: If aC credit quality is poor, price could crash

5.3 Advanced AMM Designs for Carbon Credits

Problem: Carbon credits are not fungible – a 2025 wind credit ≠ 2020 cookstove credit.

Solution: Use stratified pools or specialized curves.

Option 1: Separate Pools per Carbon Type

Pool A: aC-Wind-2025 / USDC

Pool B: aC-Solar-2025 / USDC

Pool C: aC-Cookstove-2024 / USDC

Advantage: Price differentiation

Disadvantage: Liquidity fragmentation

Option 2: Curve Finance "StableSwap" Model

For credits of similar quality (e.g., all Gold Standard, same vintage):

Formula: $A \times x \times y + x \times y = A \times k + k$

(where A is amplification coefficient)

Advantage: Lower slippage for similar assets

Disadvantage: Requires careful parameterization

Option 3: NFT-Backed Pools (C3 Protocol Approach)

- Each carbon credit = unique NFT (with metadata: project, vintage, standard)
- Pool holds basket of NFTs
- Pool token (e.g., BCT) represents claim on basket
- Users can deposit specific NFTs, withdraw generic pool tokens

Umbane Recommendation: Start with Option 1 (separate pools), transition to Option 3 as volume grows.

5.4 Liquidity Incentives ("Liquidity Mining")

Problem: New pools have no liquidity → wide spreads → no traders → still no liquidity (chicken-egg problem)

Solution: Liquidity Mining – reward LPs with additional tokens

Example: Umbane Liquidity Bootstrap

Pool: aC / USDC (on Polygon)

Target Liquidity: $500,000 (250,000 aC + $500,000 USDC)

Incentive Program:

- Reward: 50,000 UMBANE governance tokens/month
- Duration: 6 months
- Distribution: Pro-rata to LP token holders

Effective APY:

Base: 0.3% fee on $1M daily volume = $1,095/day = $400k/year = 80% APY

Bonus: 50k UMBANE tokens @ $0.50 = $25k/month = $300k/year = 60% APY

Total: 140% APY (year 1)

Precedent: KlimaDAO used KLIMA token emissions to bootstrap $1B+ in carbon credit liquidity.

5.5 Comparison: JSE Liquidity vs DeFi Liquidity

Target Market Performance: JSE Ventures vs. Umbane DeFi Pool

Metric	JSE Ventures Carbon	Umbane DeFi Pool (Target)
Liquidity Depth	R500k-R2M (estimated)	R5M-R10M (bootstrapped)
Daily Volume	R50k-R200k (episodic)	R500k-R2M (consistent)
Spreads	10-30%	0.5-3%
Trade Execution	Minutes-hours (RFQ)	Seconds (atomic swap)
Minimum Trade	R10,000+ (practical)	R100+ (fractional)
Operating Hours	09:00-17:00 SAST	24/7
API Access	Proprietary (Xpansiv)	Open (RPC calls)
Liquidity Providers	None (relies on traders)	50-100 LPs (incentivized)

6. Tokenization as Market Infrastructure

6.1 What is Tokenization?

Tokenization: Representing real-world assets (RWAs) as digital tokens on a blockchain.

For Carbon Credits:

- 1 carbon credit (1 ton CO2 offset) → 1 token on blockchain
- Token inherits credit metadata (project, vintage, standard, serial number)
- Token can be transferred, traded, retired on-chain

Benefits:

1. 24/7 Trading: Blockchain never sleeps
2. Fractionalization: 0.01 carbon credit trades are possible

3. Programmability: Smart contracts automate complex logic (e.g., auto-retirement)

4. Composability: Tokens can be used in DeFi (collateral, derivatives, etc.)

5. Transparency: All transactions public on blockchain explorer

6. Global Access: Anyone with internet can trade (no intermediary approval)

6.2 Tokenization Standards for Carbon Credits

ERC-20 (Fungible Tokens):

- Used for: Pooled/generic carbon credits

- Example: BCT (Base Carbon Tonne) = 1 token per ton CO2, any vintage/project in pool

- Advantage: High liquidity (like cash)

- Disadvantage: Loses specific credit metadata

ERC-721 (NFTs):

- Used for: Individual, unique carbon credits

- Example: Each NFT represents 1 specific credit from 1 specific project

- Advantage: Full traceability, no double-counting

- Disadvantage: Low liquidity (like collectibles)

ERC-1155 (Multi-Token):

- Used for: Batches of similar credits

- Example: 1,000 credits from same project = 1 ERC-1155 token

- Advantage: Balance between specificity and fungibility

- Disadvantage: Less common, fewer integrations

Umbane's Approach:

- mJ tokens (ERC-20): Non-Fungible energy production tokens (1 mJ = 1 Wh) (each with unique metadata)

- aC tokens (ERC-721): Fungible carbon credit tokens

- Hybrid: mJ NFTs and aC tokens can be "pooled" into ERC-20 for liquidity

6.3 Tokenization Process: Off-Chain to On-Chain

Step-by-Step (Umbane System):

STEP 1: ENERGY PRODUCTION/GENERATION

├─ Household solar panels produce electricity

├─ CT clamp sensor measures feed-in to grid

├─ PZEM-004T Module measures Voltage, Current, Power, Frequency, and Power Factor, protects the low-voltage ESP32

├─ ESP32 device records: kWh produced, timestamp, signature

├─ Smart Contract Accumulator -- software layer treats the incoming data as "drops" (millijoules) and only mints a tradeable 1 Wh token once the total reaches 3,600,000 mJ

└─ Data sent to backend via LoRaWAN

STEP 2: VERIFICATION

├─ Backend verifies device signature (tamper-proof)

├─ Cross-reference with Chainlink oracle (solar irradiance data)

├─ Optional: DAO verification committee reviews (for large claims)

└─ Approval granted

STEP 3: ENERGY TOKEN MINTING (mJ)

├─ Smart contract mints mJ tokens (1 mJ = 1 Wh)

├─ mJ tokens sent to prosumer's wallet in form of NFT

├─ On-chain record: device ID, production amount, timestamp

└─ Event emitted: EnergyRecorded(user, kwh, timestamp)

STEP 4: CARBON CREDIT CALCULATION

├─ Formula: kWh × emission factor = kg CO_2 offset

├─ Example: 1,000 kWh × 0.9 kg CO_2/kWh = 900 kg = 0.9 ton CO_2

├─ Smart contract calculates aC eligibility

├─ a prosumer who produces 1,000 kWh of solar energy would be eligible for 900 aC tokens in high carbon intensity ‘Grid Emission Factor” scenario

└─ aC token minted with metadata: project_id, vintage, amount, registry_link

STEP 5: aC TOKEN ISSUANCE (see note)

├─ aC Token sent to prosumer's wallet

├─ Token includes: project details, carbon methodology, unique serial

├─ On-chain record links to: device ID, production data, oracle attestation

└─ Event emitted: Carbon Credit issued(user, amount, nft_id)

STEP 6: TRADING / RETIREMENT

├─ Option A: Trade aC NFT on marketplace (direct peer-to-peer)

├─ Option B: Deposit aC NFT into liquidity pool, receive pool tokens

├─ Option C: Retire aC NFT on-chain (burn), claim carbon offset certificate

└─ All actions transparent on blockchain

Key Difference from I-REC:

- I-REC: Off-chain registry (Evident.app), annual batch issuance, manual processes
- Umbane aC: On-chain tokenization, instant issuance, automated processes

Key Difference from zaREC:

- zaREC: Off-chain registry via online registrar, manual processes
- Umbane aC: On-chain tokenization, instant issuance, automated processes

6.4 Preventing Double-Counting in Tokenized Systems

The Double-Counting Problem:

- Carbon credit is issued (Verra registry)
- Credit is tokenized (Toucan Protocol)
- Original credit still exists in Verra
- Risk: Both on-chain token AND off-chain credit could be used to claim offset

Solution: Registry-to-Chain Bridges with Retirement

Toucan Protocol Method (Industry Standard):

1. User holds carbon credit in Verra registry
2. User initiates bridge to blockchain
3. Toucan retires credit in Verra (marks as "used for blockchain")
4. Toucan mints equivalent TCO2 token on-chain
5. TCO2 token is 1:1 backed by retired Verra credit

6. When TCO2 is "retired" on-chain, it's burned (no longer circulating)

Result: Credit exists in only ONE place at any time (either off-chain OR on-chain, never both).

Umbane's Approach:

- aC tokens are natively issued on-chain (no prior off-chain existence)
- Each aC Token has unique serial number linked to device + production event
- To bridge to traditional registry (COAS, Verra), aC must be burned on-chain first
- Bridge smart contract emits attestation that aC was destroyed
- Traditional registry can then issue corresponding off-chain credit

Registry Integration:

OPTION 1: aC stays on-chain only (voluntary market, DeFi use)

OPTION 2: aC burned → COAS listing (for carbon tax compliance)

OPTION 3: aC burned → Verra/Gold Standard (for international recognition)

6.5 Case Studies: Successful Carbon Tokenization Projects

Toucan Protocol (2021-2026):

- Bridged: 25M+ carbon credits to Polygon
- Standards: Verra VCUs (retired from Verra, minted as TCO2)
- Pools: BCT (Base Carbon Tonne), NCT (Nature Carbon Tonne)

- Impact: Created $1B+ carbon liquidity on-chain
- Criticism: Quality concerns (some low-quality credits bridged)

KlimaDAO (2021-2026):

- Treasury: 17.5M carbon credits locked
- Token: KLIMA (backed by carbon credits)
- Mechanism: Bonding + staking + liquidity pools
- Impact: Drove carbon credit demand, but price volatility issues

C3 Protocol (2022-2026):

- Bridge: Permissionless, automated
- Standards: Verra, Gold Standard, ACR, CAR
- Innovation: Universal carbon pools (UBO, NBO)
- Advantage: Lower fees than Toucan, wider standard support

Carbonmark (2024-2026):

- Focus: API-first carbon offsetting
- Integration: Corporate ESG platforms, fintech apps
- Model: On-chain retirement with instant certificates
- Target: Developer ecosystem (Stripe for carbon)

Lessons for Umbane:

1. Start Native: Issue tokens on-chain from day 1 (don't bridge legacy credits)
2. Quality Matters: Rigorous verification prevents reputation damage
3. API-First: Make integration easy for developers/corporates

4. Incentive Alignment: Liquidity mining + DAO governance = community buy-in

7. Umbane's Technical Solution

7.1 System Architecture Overview

```
┌──────────────────────────────────────────────────────────────────────────┐
│                    PHYSICAL LAYER
│
│
│
│  Household Solar  →  CT Clamp Sensor  →  ESP32 + LoRa
│
│    (Production)      (Measurement)       (Signing + TX)
│
└──────────────────────────────────────────────────────────────────────────┘
                        ↓ LoRaWAN
┌──────────────────────────────────────────────────────────────────────────┐
│                  COMMUNICATION LAYER
│
│
│
│  LoRa Gateway  →  Backend Server  →  Chainlink Oracle
│
│   (2-15km)      (Verification)       (Price Feed + Attestation)
│
```

```
└──────────────────────────────────────────────────────────────┘

                    ↓ Signed Transactions

┌──────────────────────────────────────────────────────────────┐
│                 BLOCKCHAIN LAYER (Polygon)
│
│
│
│ Smart Contracts:
│
│ ├─ Token.sol: mJ (energy) + aC (carbon) issuance
│
│ ├─ Governor.sol: DAO governance for aC holders
│
│ ├─ Oracle.sol: Chainlink integration
│
│ └─ Marketplace.sol: P2P aC trading
│
└──────────────────────────────────────────────────────────────┘

                    ↓

┌──────────────────────────────────────────────────────────┐
│                 DEFI LAYER
│
│
│
│ Liquidity Pools (Uniswap V3 on Polygon):
│
│ ├─ aC / USDC pool (primary trading pair)
│
│ ├─ mJ / MATIC pool (energy token speculation)
```

```
│  └─ aC / KLIMA pool (bridge to KlimaDAO)
│
│
│  Yield Strategies:
│
│  ├─ LP staking (earn UMBANE governance tokens)
│  ├─ aC staking (lock for governance rights)
│  └─ Automated vaults (Yearn-style optimizers)
└──────────────────────────────────────────────────────────────┘
                                ↓
┌──────────────────────────────────────────────────────────────┐
│                    BRIDGE LAYER (Optional)
│
│  aC Token Burn  →  Bridge Contract  →  Off-Chain Registry
│   (Destroy token)   (Attestation)       (COAS/Verra/GoldStd)
│
│
│  Use Cases:
│
│  ├─ Submit to COAS (carbon tax compliance)
│  ├─ Submit to JSE (traditional market)
│  └─ Submit to Verra (international recognition)
└──────────────────────────────────────────────────────────────┘
```

Here is a Mermaid JS chart:

```
graph TD
    subgraph Physical_Layer [Physical Layer]
    A[Household Solar] -->|Energy| B[CT Clamp Sensor]
    B -->|Current Data| C[ESP32 + LoRa]
    end

    C -->|LoRaWAN 2-15km| D[LoRa Gateway]

    subgraph Communication_Layer [Communication Layer]
    D --> E[Backend Server]
    E <--> F[Chainlink Oracle]
    end

    E -->|Signed Transactions| G[Blockchain Layer: Polygon]

    subgraph Blockchain_Layer [Blockchain Layer]
    G --> H[Token.sol: mJ + aC Issuance]
    G --> I[Governor.sol: DAO]
    G --> J[Oracle.sol: Data Feed]
    G --> K[Marketplace.sol: P2P]
    end

    H --> L[DeFi Layer]
```

```
subgraph DeFi_Layer [DeFi Layer]
L --> M[Uniswap V3: aC/USDC, mJ/MATIC]
L --> N[Staking & Yield Vaults]
end

N --> O[Bridge Layer: Optional]

subgraph Bridge_Layer [Bridge Layer]
O --> P[Token Burn]
P --> Q[Bridge Contract]
Q --> R[Off-Chain Registries: Verra/COAS]
end
```

7.2 Token Economics (Do we mint an NFT?)

Two-Token Model:

mJ Token (Energy Production Token):

- Type: ERC-20 (fungible)
- Supply: Unlimited (minted based on verified production)
- Unit: 1 mJ = 1 Wh (watt-hour) of energy produced
- Purpose:
  - Proof of energy production
  - Intermediate step before aC issuance
  - Can be traded independently (energy futures market)

- Minting Trigger: IoT device measurement → oracle verification → smart contract mint

- Example: 1,000 kWh production = 1,000,000 mJ tokens minted

aC Token (Carbon Credit NFT):

- Type: ERC-721 (NFT, unique)

- Supply: Tied to verified mJ conversion (based on emission factor)

- Unit: 1 aC NFT = X kg CO_2 offset (variable by project)

- Purpose:

  - Carbon credit for trading

  - Governance rights (DAO voting)

  - Retirement for carbon offsetting

- Metadata: Project ID, vintage, emission factor, device ID, production period

- Minting Trigger: Accumulated mJ → conversion formula → DAO verification → aC NFT mint

- Example: 3,600 mJ (1,000 kWh) × 0.9 kg CO_2/kWh = 900 kg CO_2 → aC NFT #12345 (0.9 tons)

Conversion Formula:

aC Amount (kg CO_2) = mJ tokens × kWh conversion × Emission Factor

Where:

- mJ tokens = raw energy production (1 mJ = 1 Wh)

- kWh conversion = mJ / 1,000,000 (convert to kilowatt-hours)

- Emission Factor = kg CO_2 offset per kWh (SA grid: ~0.9 kg/kWh as of 2026)

Example:

10,000 kWh solar production

= 10,000,000 mJ tokens

= 10,000 kWh × 0.9 kg CO_2/kWh

= 9,000 kg CO_2 offset

= 9 aC NFTs (assuming 1 NFT = 1 ton CO_2)

aC Pricing Mechanisms:

1. Primary Market (Newly Minted aC):
   - Algorithmic floor price: Max(JSE carbon spot × 0.8, R100/ton)
   - Example: If JSE VCU = R150, floor = R120/ton
   - Minting cost: R50/aC (covers oracle + tx costs)
   - Prosumer receives: R120 - R50 = R70 net per ton

2. Secondary Market (aC Trading):
   - DeFi Pool: aC/USDC on Uniswap (market price)
   - P2P: Direct wallet-to-wallet trades
   - Marketplace: Umbane marketplace with offers/bids
   - Price discovery via: AMM curve + arbitrage

3. Bridge Price (aC → Off-Chain):
   - Bridge fee: 5% of off-chain value
   - Example: aC = R120, COAS credit = R180 → bridge fee = R9

- Net to prosumer: R180 - R9 = R171 (if they choose to bridge)

7.3 Liquidity Pool Design

Primary Pool: aC / USDC (Uniswap V3, Polygon)

Initial Liquidity:
- 100,000 aC Tokens → pooled as ERC-20 "aC-Pool" token
- $175,000 USDC (at $1.75/aC)
- Concentrated liquidity: $1.50-$2.00 range (Uniswap V3 feature)
- Fee tier: 0.3% (standard for volatile pairs)

Liquidity Bootstrap Program:
- Phase 1 (Months 1-3): 100% UMBANE token rewards to LPs
  - 50,000 UMBANE/month (decreasing emissions schedule)
  - Target: $500k TVL (Total Value Locked)
- Phase 2 (Months 4-12): 50% UMBANE + 50% aC yield farming
  - LPs stake LP tokens → earn UMBANE + yield from aC appreciation
- Phase 3 (Year 2+): Protocol fees + DAO treasury support
- Sustainable via 0.05% protocol fee on swaps

Liquidity Provider Returns:

Scenario: Alice provides $10,000 liquidity (5,000 aC + $10,000 USDC)

Year 1 Earnings:

├─ Trading Fees: 0.3% × $2M annual volume × 2% pool share = $1,200

├─ UMBANE Rewards: 50,000 tokens × 2% share × $0.50/token = $6,000

├─ aC Price Appreciation: 5,000 aC × ($2.00 - $1.75) = $1,250

└─ Total: $8,450 on $10,000 = 84.5% APY

Risks:

├─ Impermanent Loss: If aC price drops to $1.50, IL = ~5%

├─ Smart Contract Risk: Bug in Uniswap/Umbane contracts

└─ aC Quality Risk: If credits not recognized, price crash

7.4 Price Discovery Mechanism

Multi-Source Price Aggregation:

```
┌──────────────────────────────────────────────┐
│              UMBANE PRICE ORACLE
│
│
│
│  Input Sources (weighted average):
│   ├─ 60%: Uniswap aC/USDC pool (on-chain, real-time)
│   ├─ 20%: Chainlink JSE carbon price feed (if available)
```

```
│  ├─ 10%: Xpansiv/CBL spot price (via API, if accessible)
│  └─ 10%: TWAP (Time-Weighted Average Price, 24hr)
│
│
│  Output: aC Reference Price (updated every 10 minutes)
│  └─ Used for: DAO treasury, collateral, derivatives
└──────────────────────────────────────────────────────────┘
```

API Access:

```
// Public API endpoint (no auth required)
GET https://api.umbane.co.za/v1/price/aC

Response:
{
  "price_zar": 178.50,
  "price_usd": 9.92,
  "volume_24h": 50000,
  "liquidity_tvl": 875000,
  "sources": {
    "uniswap": 180.00,
    "jse_feed": 175.00,
    "xpansiv": 177.00,
    "twap_24h": 178.25
  },
  "timestamp": "2026-03-20T14:35:00Z"
}
```

Comparison to JSE/Xpansiv:

- JSE: No public API, prices via Bloomberg terminal or broker (R50k/year subscription)

- Umbane: Open API, free, real-time updates every 10 min

7.5 DAO Governance Integration discussion

aC Token = Governance Rights:

- 1 aC Token staked = 1 vote in DAO

- Voting on:

  - Carbon price parameters (emission factor adjustments)

  - Device verification standards

  - Treasury allocation (liquidity incentives, grants)

  - Protocol upgrades (smart contract changes)

- Bridge partnerships (which registries to support)

Governance Process:

1. PROPOSAL CREATION

   ├─ Any aC holder with >100 aC can propose

   ├─ Proposal template: Title, Description, On-chain actions

   └─ Submission fee: 10 aC (refunded if passes)

2. DISCUSSION PERIOD (5 days)

   ├─ Community debate on Discord/Forum

├─ Proponent can amend proposal

└─ "Signal vote" on Snapshot (gasless)

3. VOTING PERIOD (7 days)

├─ On-chain vote via Governor contract

├─ Quorum: 4% of total aC staked

├─ Threshold: 51% for standard, 67% for protocol upgrades

└─ Voting power = aC staked (linear, no quadratic)

4. TIMELOCK (48 hours)

├─ If passed, proposal enters timelock

├─ Safety period for users to exit if they disagree

└─ Emergency cancel by multi-sig (3/5 founders)

5. EXECUTION

├─ Anyone can trigger execution after timelock

├─ Smart contracts automatically implement changes

└─ Results published on-chain + off-chain

Example Proposals:

- AIP-001: Adjust emission factor from 0.9 to 0.85 kg CO_2/kWh (based on Eskom grid changes)

- AIP-002: Allocate R500k from treasury to deploy 10 more LoRa gateways in Stellenbosch

- AIP-003: Partner with Gold Standard for aC certification pathway

7.6 Integration with Traditional Markets (JSE Bridge)

Bridge Use Case: Prosumer wants to sell aC on JSE Ventures for higher price (or tax compliance).

Process:

STEP 1: aC HOLDER INITIATES BRIDGE

├─ User connects wallet to Umbane Bridge interface

├─ Selects mJ NFTs to bridge (e.g., 10 aC = 10 tons CO2)

└─ Confirms transaction (pays gas + bridge fee)

STEP 2: ON-CHAIN BURN

├─ Bridge contract burns NFTs (permanent destruction)

├─ Event emitted: BridgeToCOAS(user, aC_ids[], timestamp)

├─ Merkle proof generated (on-chain record of burn)

└─ User receives bridge certificate (signed by bridge oracle)

STEP 3: OFF-CHAIN SUBMISSION

├─ User submits bridge cert to COAS registry

├─ Documents include:

│ ├─ Merkle proof of mJ/aC burn

│ ├─ Device production data (from blockchain)

│ ├─ Carbon methodology (Umbane standard document)

│ └─ Verification attestation (Chainlink oracle + DAO vote)

├─ COAS reviews submission (2-4 weeks)

└─ If approved, COAS issues equivalent off-chain credits

STEP 4: JSE TRADING

├─ User lists COAS credits on JSE Ventures (via Xpansiv)

├─ Trades on traditional order book

├─ Settlement: Credits transferred to buyer, user receives ZAR

└─ Credits can be used for carbon tax compliance

BRIDGE ECONOMICS:

├─ aC value on Umbane: R150/ton

├─ COAS credit value on JSE: R180/ton

├─ Bridge fee: 5% × R180 = R9

├─ Net to user: R180 - R9 = R171

├─ Arbitrage profit: R171 - R150 = R21/ton (12% gain)

└─ Bridge demand creates upward pressure on aC price

Reverse Bridge (JSE → Umbane):

- Not initially supported (to avoid quality issues from unknown projects)
- Could be added via DAO vote if:
  - Verra/Gold Standard credits only
  - SA-based projects only
  - Vintage restrictions (post-2024)
- 1:1 burn ratio (Verra credit retired before aC minted)

8. Bridging TradFi and DeFi Carbon Markets

8.1 The Two Parallel Markets

TradFi Carbon Market (JSE, COAS, Verra):

- Centralized registries
- Manual processes (weeks for credit issuance)
- High barriers to entry (I-REC registration, consultants)
- Limited API access (proprietary systems)
- Business hours trading
- Regulatory recognition (carbon tax eligibility)

DeFi Carbon Market (Umbane, Toucan, KlimaDAO):

- Decentralized (blockchain)
- Automated (instant issuance)
- Low barriers (wallet + internet)
- Open APIs (smart contracts)
- 24/7 trading
- Regulatory ambiguity (not yet recognized for tax)

8.2 Bridge Value Proposition

Why Bridge?

For Prosumers (aC → COAS/JSE):

1. Price Arbitrage: COAS credits trade at premium to DeFi (~20%)
2. Tax Compliance: Corporate buyers need COAS credits for carbon tax

3. Legitimacy: Some corporates don't trust "crypto" credits yet

4. Liquidity Access: JSE has different buyer base (institutions)

For Corporates (COAS → aC):

1. Cost Savings: DeFi credits are cheaper (~15-20% discount)

2. Instant Settlement: No waiting for registry processes

3. Fractional Purchases: Can buy 0.5 tons (not possible in TradFi)

4. Transparency: Full on-chain audit trail

8.3 Technical Bridge Architecture

Option 1: Custodial Bridge (Umbane Operated)

```
┌──────────────────────────────────────────────────────────┐
│                UMBANE BRIDGE OPERATOR
│
│  ┌──────────────────┐        ┌──────────────────────┐
│  │ DeFi Wallet      │        │ TradFi Account       │
│  │ (Polygon)        │        │ (COAS/Verra)         │
│  └──────────────────┘        └──────────────────────┘
│       ↓                            ↓
│   aC Burn Event            Credit Submission
│       ↓                            ↓
│  Verification ←──────────────────→ Manual Process
│       ↓                            ↓
│   Bridge Fee               Off-Chain Credit
```

```
│      Collected                Issued
└──────────────────────────────────────────────────────┘
```

Advantages:

- Simple for users (one-click bridge)
- Fast (2-4 weeks vs 3-6 months DIY)
- Quality control (Umbane vets all submissions)

Disadvantages:

- Trust required (Umbane is custodian)
- Single point of failure (if Umbane shuts down)
- Regulatory risk (Umbane liable for bridge quality)

Option 2: Decentralized Bridge (DAO Operated)

```
┌──────────────────────────────────────────┐
│                DAO BRIDGE COMMITTEE
│            (5 elected aC holders)
│
│  User submits aC burn → Committee reviews →
│  3/5 multi-sig approval → Bridge attestation issued →
│  User submits to COAS independently
└──────────────────────────────────────────┘
```

Advantages:

- Decentralized (no single party control)
- Transparent (all approvals on-chain)
- Censorship-resistant (DAO decides, not company)

Disadvantages:

- Slower (committee review takes time)
- Higher complexity for users (more steps)
- Coordination overhead (committee compensation, turnover)

Umbane Recommendation: Start with Option 1 (custodial, fast), transition to Option 2 after 12-18 months once DAO matures.

8.4 Bridge Economics and Incentives

Bridge Fee Structure:

aC → COAS Bridge:

├─ Base Fee: 5% of COAS value

├─ Verification Fee: R50/ton (covers oracle + committee)

├─ Total Fee: 5% + R50

Example:

├─ aC value (DeFi): R150/ton

├─ COAS value (TradFi): R180/ton

├─ Bridge Fee: (5% × R180) + R50 = R9 + R50 = R59

├─ Net to user: R180 - R59 = R121

└─ Still profitable vs selling aC at R150 (R121 > R150? No!)

Adjustment:

├─ Reduce verification fee to R20/ton (automation)

├─ Bridge Fee: R9 + R20 = R29

├─ Net to user: R180 - R29 = R151

└─ Profit: R151 - R150 = R1/ton (marginal, but works)

OR

├─ Wait for aC price to appreciate to R160 (via liquidity)

├─ Bridge Fee: R29

├─ Net to user: R180 - R29 = R151

└─ Profit: R151 - R160 = -R9 loss (bridge not economical)

Conclusion: Bridge only economical when COAS premium > (Bridge Fee / aC price)

Bridge Flow Dynamics:

- If COAS premium high (>20%) → aC holders bridge → aC supply decreases → aC price rises

- If COAS premium low (<10%) → No bridging → aC stays on-chain → DeFi liquidity grows

- Equilibrium: aC price converges to ~90% of COAS (10% bridge fee + friction)

8.5 Regulatory Recognition Strategy

Goal: Get aC tokens recognized as eligible for SA Carbon Tax compliance

Pathway:

PHASE 1: Build Track Record (Year 1-2)

├─ Deploy 1,000+ devices in Cape Town

├─ Issue 50,000+ aC EC20 / mJ NFTs (50,000 tons CO2)

├─ Establish DAO governance (transparent, auditable)

├─ Partner with local verifier (e.g., SouthSouthNorth)

└─ Document methodology (peer-reviewed carbon accounting)

PHASE 2: Engage Regulators (Year 2-3)

├─ Present to National Treasury (Carbon Tax unit)

├─ Demonstrate: Quality, transparency, scale, anti-fraud measures

├─ Propose: "Digital Carbon Credit" category in COAS

├─ Show: Cost savings for government (automated vs manual verification)

└─ Pilot: 10 companies use aC for partial tax compliance (with Treasury approval)

PHASE 3: Legislative Change (Year 3-5)

├─ Advocate for Carbon Tax Act amendment

├─ New eligible category: "Blockchain-verified domestic solar offsets"

├─ Requirements: DAO governance, oracle verification, open-source code

├─ aC meets all requirements → eligible for COAS listing

└─ Result: aC holders can submit directly to COAS (no bridge needed)

FALLBACK: Voluntary Market Focus

├─ If regulatory path fails, focus on voluntary commitments

├─ Corporates with net-zero targets (not carbon tax compliance)

├─ Examples: Nedbank, Woolworths, Pick n Pay, Capitec

└─ Pitch: Cheaper, transparent, supports local prosumers

9. Regulatory Pathways and Compliance

9.1 Current Regulatory Landscape

Carbon Tax Act (2019, updated 2024):

- aC tokens: Not explicitly recognized as offsets
- COAS eligibility: Requires credits from Verra, Gold Standard, or CDM
- Gap: No provision for "natively digital" carbon credits

Financial Sector Conduct Authority (FSCA):

- Crypto Asset Regulatory Framework (2026) (check)
- aC tokens may be classified as "crypto asset"
- Implication: Umbane may need FSCA registration if aC is "financial product"
- Exemption: If aC is "utility token" (not investment), may be exempt

ICASA (Radio Spectrum):

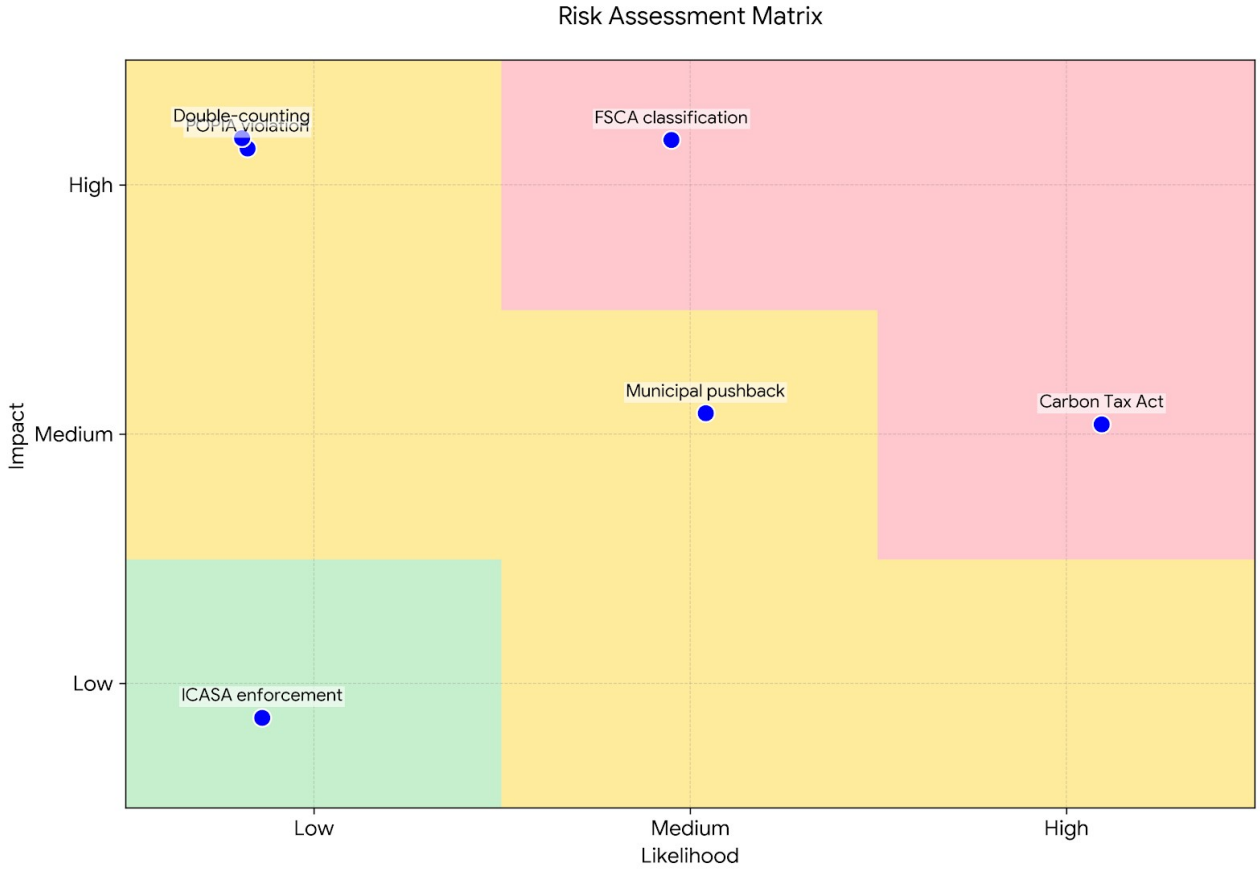
- LoRa devices (868MHz) fall under "short-range devices"
- Exempt from licensing if <10mW, <10% duty cycle

- Action: Obtain Type Approval for ESP32 LoRa devices

POPIA (Data Protection):

- User energy data is "personal information"
- Umbane must: Obtain consent, secure storage, allow access/deletion
- Compliance: Privacy policy, data encryption, GDPR-style rights

9.2 Regulatory Risk Matrix



Risk Assessment and Mitigation Strategy

Risk	Likelihood	Impact	Mitigation Strategy
FSCA classification as financial product	Medium	High (requires licensing)	Structure aC as "utility token" with clear use case (carbon offsetting)
Carbon Tax Act non-recognition	High	Medium (limits corporate demand)	Focus on voluntary market first, engage Treasury early
ICASA enforcement on LoRa devices	Low	Low (fines, not shutdown)	Obtain Type Approval, limit power to <10mW
POPIA violation (data breach)	Low	High (fines + reputation)	Encrypt all data, conduct POPIA audit
Municipal pushback (CoCT)	Medium	Medium (PR damage)	Partner with City on pilot (show alignment with solar goals)
Double-counting accusations	Low	High (destroys credibility)	Implement robust on-chain serial numbers, open-source verification

9.3 Proactive Regulatory Engagement Plan

Month 1-3: Research & Documentation

- Hire carbon accounting consultant (e.g., SouthSouthNorth, Climate Neutral Group SA)
- Draft: Carbon Methodology Document (peer-reviewed)
- Draft: Legal opinion on aC token classification (FSCA compliance)
- Draft: POPIA compliance policy

Month 4-6: Stakeholder Outreach

- National Treasury (Carbon Tax unit): Present Umbane concept
- Department of Mineral Resources & Energy (DMRE): COAS administrators
- City of Cape Town (Energy Directorate): Feed-in tariff alignment
- Solar PV Industry Association (SAPVIA): Industry endorsement

Month 7-12: Pilot with Approval

- Apply for "Regulatory Sandbox" (if available via FSCA)
- Pilot: 100 households, voluntary carbon market only (no tax claims)
- Quarterly reporting to regulators (transparency builds trust)
- Publish: Open-source code, audit reports, carbon accounting methodology

Year 2: Scale & Legitimacy

- Expand to 1,000 households
- Partner with university (UCT, Stellenbosch) for academic validation
- Submit: Peer-reviewed paper on "Blockchain Carbon Credits in SA"
- Apply: Gold Standard or VCS methodology approval for aC

Year 3: Legislative Advocacy

- Join: Carbon Pricing Coalition (industry group)
- Lobby: Carbon Tax Act amendment to recognize digital credits
- Propose: "Blockchain Offset" category in COAS regulations
- Goal: aC eligible for carbon tax compliance by 2029

9.4 Compliance Checklist for Launch

Legal:

- [ ] Company registration (Umbane NPC or Pty Ltd)
- [ ] FSCA legal opinion on aC token status
- [ ] POPIA compliance audit + policy
- [ ] Terms of Service + Privacy Policy (published on website)

Technical:

- [ ] Smart contract security audit (OpenZeppelin, Certik)
- [ ] Penetration testing (backend API, database)
- [ ] Data encryption (AES-256, TLS 1.3)
- [ ] Disaster recovery plan (backup keys, data recovery)

Carbon Accounting:

- [ ] Methodology document (peer-reviewed by carbon consultant)
- [ ] Emission factor justification (based on Eskom/DMRE data)
- [ ] Additionality demonstration (solar production wouldn't happen without aC incentive)
- [ ] Baseline scenario (grid electricity displaced)

Operational:

- [ ] Device Type Approval (ICASA, if required)
- [ ] Insurance (liability for device malfunction, cyber insurance)
- [ ] Customer support system (Discord, email, phone)
- [ ] Fraud reporting mechanism (bug bounty, whistleblower process)

10. Economic Modeling and Impact Projections

10.1 Market Opportunity Analysis

Addressable Market (Cape Town Metro):

- Total households: ~1.2 million
- Households with solar: ~50,000 (4% penetration, growing 20%/year)

- Avg. solar system size: 5 kW
- Avg. production: 8,000 kWh/year
- Carbon offset potential: 8,000 × 0.9 kg/kWh = 7.2 tons CO_2/year

Total Addressable Market (TAM):

- 50,000 households × 7.2 tons/year = 360,000 tons CO_2/year
- At R150/ton (aC DeFi price) = R54M/year ($3M USD)
- At R180/ton (COAS price) = R64.8M/year ($3.6M USD)

Serviceable Addressable Market (SAM) (Umbane captures 20% in 5 years):

- 10,000 households enrolled
- 72,000 tons CO_2/year
- R10.8M/year revenue potential ($600k USD)

10.2 Revenue Model

Umbane Revenue Streams:

1. Device Sales: R2,100/device × 10,000 = R21M one-time
2. Platform Fee: 5% of aC trades
   - Year 1: R1M aC trading volume × 5% = R50k
   - Year 5: R20M volume × 5% = R1M
3. Bridge Fee: 5% of COAS bridging
   - Year 3: R5M bridged × 5% = R250k

4. API Licensing: Corporates integrating Umbane API

- R5,000/month per client × 10 clients = R600k/year

5. Staking Rewards: 2% of DAO treasury (community owned)

- Not direct revenue, but incentivizes governance

Total Revenue Projection (5 years):

Year 1: R21M (devices) + R200k (platform fees) = R21.2M

Year 2: R15M (devices) + R500k (fees) = R15.5M

Year 3: R10M (devices) + R1.5M (fees) + R250k (bridge) = R11.75M

Year 4: R8M (devices) + R3M (fees) + R500k (bridge) + R600k (API) = R12.1M

Year 5: R5M (devices) + R5M (fees) + R1M (bridge) + R1.2M (API) = R12.2M

5-Year Total: R72.75M (~$4M USD)

Cost Structure:

- Device Manufacturing: R1,500/unit (70% of sale price)
- Backend Infrastructure: R200k/year (servers, oracles)
- Development Team: R3M/year (5 devs @ R600k)
- Marketing: R1M/year
- Legal/Compliance: R500k/year
- Total Opex: R4.7M/year

Break-Even: Year 2 (cumulative revenue > cumulative costs)

10.3 Impact on SA Carbon Market Liquidity

Scenario Modeling:

Baseline (No Umbane):

- JSE carbon trading volume: ~R50M/year (2026)
- Spread: 15-30%
- Participation: 200 active traders
- Liquidity depth: R2M (buy/sell within 5%)

Scenario 1: Umbane Small Success (1,000 users by Year 3)

- Additional volume: R5M/year (aC trading on DeFi)
- Spillover to JSE: R1M/year (via bridge)
- Spread reduction: 15% → 12% (more price competition)
- New participants: +500 (household prosumers)

Scenario 2: Umbane Moderate Success (10,000 users by Year 5)

- Additional volume: R50M/year (matching JSE current volume!)
- Spillover to JSE: R10M/year
- Spread reduction: 15% → 8% (significant liquidity improvement)
- New participants: +10,000
- Outcome: SA carbon market volume doubles, prices stabilize

Scenario 3: Umbane Large Success (50,000 users by Year 7, post-regulatory approval)

- Additional volume: R250M/year
- Spillover to JSE: R50M/year (JSE volume +100%)
- Spread reduction: 15% → 3-5% (liquid market achieved)
- New participants: +50,000
- Outcome: SA becomes African carbon trading hub

10.4 Price Discovery Impact

Current State (JSE Ventures):

- Price updates: Every 3-5 days (when trades occur)
- Bid-ask spread: R20-50 (on R150 base = 13-33% spread)
- No public API for real-time prices

With Umbane DeFi Pool:

- Price updates: Every 2 seconds (every block)
- Bid-ask spread: R1-5 (on R150 base = 0.7-3.3% spread)
- Public API: Free, real-time

Arbitrage Mechanism:

Scenario: aC trading at R160 on Umbane, VCU at R180 on JSE

Arbitrageur Action:

1. Buy 100 aC on Umbane = R16,000

2. Bridge aC → COAS (pay R29/ton fee) = R2,900

3. Sell COAS on JSE = R18,000

4. Profit: R18,000 - R16,000 - R2,900 = -R900 (loss!)

Conclusion: Spread must be >bridge fee for arbitrage

When does arbitrage work?

├─ JSE price - Umbane price > Bridge fee (R29/ton)

├─ R180 - R150 = R30 > R29 ✓ (profitable by R1/ton)

└─ Result: Arbitrage keeps prices aligned within R30 band

Impact on JSE:

- Umbane acts as "price floor" for JSE carbon credits

- If JSE price drops below (Umbane aC + Bridge fee), arbitrage restores equilibrium

- Result: More stable JSE pricing, fewer wild swings

11. Recommendations and Next Steps

11.1 For Umbane: Strategic Priorities

Immediate (Next 3 Months):

1. Regulatory Clarification:

   - Obtain FSCA legal opinion on aC token status (R50k)

   - Draft Carbon Methodology Document (hire consultant, R100k)

- Apply for ICASA Type Approval for LoRa devices (R20k)

2. Technology Validation:

- Deploy 10-device pilot in Cape Town (existing project location)
- Test end-to-end: CT clamp → LoRa → backend → Polygon mint
- Security audit on smart contracts (R200k)

3. Stakeholder Engagement:

- Meet with Chris Sturgess (JSE) to present Umbane as complementary to JSE (not competitor)
- Meet with National Treasury (Carbon Tax unit) to discuss regulatory pathway
- Partner with local solar installer (Solid Solar, Solar Advice) for device distribution

Short-Term (Months 4-12):

1. Community Bootstrap:

- Launch: 100-household pilot in Cape Town
- Deploy: 3 LoRa gateways (Muizenberg, Fish Hoek, Kalk Bay)
- Create: aC/USDC liquidity pool with R500k seed liquidity

2. API Development:

- Build: Public API for aC price feeds (real-time)
- Integration: Sample code for corporates to auto-purchase aC offsets
- Documentation: Developer portal (docs.umbane.co.za)

3. Bridge Prototype:

- Test: Manual aC → COAS bridge with 1 pilot user
- Document: Full bridge process (legal, technical, timing)
- Calculate: True bridge costs (accounting for labor, fees)

Medium-Term (Year 2-3):

1. Scale to 1,000 Households:
   - Manufacturing: Contract local PCB assembler (reduce device cost to R1,800)
   - Funding: Token sale (UMBANE governance token, R5M raise)
   - Expansion: Beyond Cape Town (Stellenbosch, Paarl, Hermanus)

2. DAO Launch:
   - Deploy: Governor contracts (OpenZeppelin)
   - Transfer: Ownership of Token.sol to Timelock (DAO controlled)
   - First vote: Emission factor parameter (community engagement)

3. Bridge Automation:
   - Develop: Self-service bridge portal (user submits aC burn → system generates COAS package)
   - Partner: Carbon consultant (e.g., SouthSouthNorth) to handle COAS submissions in bulk
   - Target: <5% bridge fee (competitive with market)

Long-Term (Year 4-5):

1. Regulatory Approval:
   - Goal: aC eligible for Carbon Tax Act compliance
   - Strategy: Peer-reviewed methodology, academic validation, Treasury

pilot

- Fallback: Focus on voluntary market (corporates with net-zero commitments)

2. Market Maker Program:

- Partner: Traditional market maker (e.g., Rand Merchant Bank) to provide JSE-Umbane arbitrage
- Incentive: DAO treasury supports MM with aC loans (0% interest)
- Result: Tighter spreads between DeFi and TradFi

3. Pan-African Expansion:

- Replicate: Umbane model in Kenya (launching national carbon registry)
- Replicate: Nigeria (large solar market, weak carbon infrastructure)
- Vision: Umbane as African carbon credit infrastructure layer

11.2 For JSE: Leverage Umbane as Liquidity Source

Problem: JSE faces chronic liquidity shortage in carbon market

Opportunity: Umbane brings new supply (household solar) AND new demand (DeFi traders)

Recommendations for JSE:

1. Recognize Umbane aC as "Pre-COAS" Assets:

- Create new trading category on JSE Ventures: "Digital Carbon Credits (Pending COAS)"
- Allow aC tokens to be listed with disclaimer ("not yet tax-eligible")

- Provide pathway: aC → COAS conversion on JSE platform itself

2. Open API Partnership:

- Work with Xpansiv or zaRIC to provide public read-only API for JSE carbon prices

- Umbane integrates JSE prices into aC oracle → aligns markets

- Result: JSE becomes benchmark for all SA carbon markets (on-chain + off-chain)

3. Liquidity Pool Co-Investment:

- JSE (or JSE member firms) provide seed capital to aC/USDC pool (R1M-R2M)

- Earn: LP fees + UMBANE tokens

- Benefit: Deeper liquidity → arbitrage between JSE and Umbane → tighter spreads on JSE

4. Market Maker Recruitment:

- JSE incentivizes member firms to act as market makers (rebates on trading fees)

- Market makers arbitrage: Buy aC on Umbane (low price) → Bridge → Sell on JSE (high price)

- Result: Continuous price discovery, reduced volatility

11.3 For National Treasury: Regulatory Sandbox for Digital Carbon

Problem: Current Carbon Tax Act doesn't address blockchain-based credits

Opportunity: South Africa can be global leader in digital carbon credit

regulation

Recommendations:

1. Create "Digital Carbon Credit" Category in COAS:

   - Criteria: Blockchain-based, open-source verification, DAO governance

   - Requirements: Annual audit, methodology peer-review, serial number tracking

   - Pilot: 10 companies use digital credits for max 5% of carbon tax liability (2027-2028)

2. Regulatory Sandbox:

   - Allow Umbane (and similar projects) to operate under "test regime"

   - 3-year pilot: Monitor for fraud, double-counting, quality issues

   - If successful: Full recognition in Carbon Tax Act amendment (2029)

3. I-REC/D-REC Reform:

   - Designate local South African D-REC issuer

   - Lower barriers: Household solar eligible for D-REC (sliding fee scale)

   - Result: Domestic prosumers can participate in formal REC market

11.4 For Household Prosumers: Why Participate in Umbane?

Benefits:

1. Earn Extra Income:

   - Current: City pays R0.75/kWh feed-in tariff

   - Umbane: Additional R0.10-R0.15/kWh in aC value

- Example: 5 kW system producing 8,000 kWh/year
  - City feed-in: R6,000/year
  - aC tokens: R1,200/year
  - Total: R7,200/year (+20% income boost)

2. Support Community:
   - DAO voting rights (shape future of Cape Town's energy system)
   - Community projects funded by DAO treasury (more LoRa gateways, education)

3. Climate Impact:
   - Verified carbon offsets (transparent, on-chain)
   - Support other prosumers (liquidity pools benefit everyone)

4. Simple Setup:
   - One-time device installation (2 hours, electrician not required for CT clamp)
   - Automatic token minting (no paperwork)
   - Trade on phone (MetaMask or Umbane mobile app)

Risks:

- New technology (smart contracts could have bugs)
- Regulatory uncertainty (aC may never be tax-eligible)
- Price volatility (aC value could fluctuate ±30% monthly)

Decision Framework: Participate if:

- You have solar (3+ kW system)

- You're comfortable with crypto (or willing to learn)
- You believe in decentralization (vs City monopoly)

11.5 For Corporates: Why Buy aC Tokens vs Traditional Credits?

Advantages of aC:

1. Cost: 15-20% cheaper than COAS/Verra credits
2. Transparency: Full audit trail on blockchain (ESG reporting)
3. Speed: Instant purchase via API (no broker, no weeks of waiting)
4. Local Impact: Support Cape Town households (vs international offsets)
5. Innovation: Position company as "blockchain-forward" (PR value)

Disadvantages:

- Not (yet) eligible for carbon tax compliance
- Regulatory risk (if Treasury doesn't recognize, value could drop)
- Crypto complexity (finance team needs to learn wallet management)

Ideal Corporate Profiles:

- Tech Companies: Comfortable with crypto (Takealot, Luno, Yoco)
- Retailers with Net-Zero Goals: Woolworths, Pick n Pay, Capitec
- Large Banks: Nedbank (already buys Credible Carbon), Standard Bank, FirstRand
- Solar Companies: Solareff, SunExchange (buy aC, bundle with installations)

API Integration Example:

```
// Corporate auto-offsets monthly emissions
```

```
const umbane = require('umbane-sdk');

async function offsetMonthlyEmissions() {
  const emissions_kg = await fetchCompanyEmissions(); // e.g., 50,000 kg CO2
  const aC_needed = emissions_kg / 1000; // Convert to tons

  const quote = await umbane.getQuote({
    amount: aC_needed,
    delivery: 'instant'
  });

  const purchase = await umbane.buy({
    amount: aC_needed,
    price_limit: quote.price * 1.05, // 5% slippage tolerance
    wallet: process.env.COMPANY_WALLET
  });

  await umbane.retire({
    aC_ids: purchase.token_ids,
    reason: 'Monthly emissions offset - March 2026'
  });

  console.log(`Offset ${aC_needed} tons CO2 for R${purchase.total}`);
  return purchase.certificate_url; // For ESG reporting
}
```

12. Conclusion: Breaking the Logjam

The Core Problem (Revisited)

Chris Sturgess (JSE) identified the pain points:

1. Carbon credits "don't appear to be easily tradeable"
2. Prices "all over the place"
3. Single-vendor dependency (Xpansiv/Evident)
4. No public API/SDK for integration
5. High barriers (I-REC registration via GCC Sheffield)

The Root Cause

The SA carbon market suffers from infrastructure poverty:

- No market makers → thin markets
- No continuous trading → stale prices
- No public APIs → developer lock-out
- High entry costs → small producers excluded
- Centralized bottlenecks → inertia and friction

This is a classic market failure – all the pieces exist (regulation, demand, projects), but they can't connect efficiently.

The Umbane Solution

Tokenization + DeFi unlocks infrastructure:

1. Automated Market Makers → continuous liquidity (0.3% spreads vs 15-30%)

2. 24/7 Trading → real-time price discovery

3. Open APIs → anyone can integrate (free, no vendor lock-in)

4. Low Barriers → R1,200 device vs R25k I-REC registration

5. Decentralized → no single point of failure (DAO governance)

The Bigger Vision

Umbane is not just about Cape Town solar prosumers. It's about democratizing carbon markets across Africa:

- Phase 1: Cape Town (10,000 households)

- Phase 2: South Africa (50,000 households)

- Phase 3: Pan-African (500,000 households)

If successful, Umbane becomes the Uniswap of carbon credits – open, permissionless, liquid.

Call to Action

For JSE: Partner with Umbane to bring DeFi liquidity to TradFi carbon markets. Don't see blockchain as a threat – see it as infrastructure that makes your market better.

For National Treasury: Recognize digital carbon credits in regulatory frameworks. South Africa can lead Africa (and the world) in blockchain carbon governance.

For Household Prosumers: Join the pilot. Your solar panels can do more than reduce your electricity bill – they can help price discovery for an entire market.

For Developers: Umbane's open APIs and smart contracts are your playground. Build the next generation of carbon-aware applications.

Final Thought

Markets need liquidity. Liquidity needs infrastructure. Infrastructure needs openness.

We believe the future of carbon markets is open.

We believe the future of energy markets is onchain

Appendices

Appendix A: Glossary of Terms

- aC Token: Umbane's carbon credit token (1 aC = X kg CO2 offset)
- AMM: Automated Market Maker (algorithm that provides liquidity)
- BCT: Base Carbon Tonne (Toucan Protocol's pooled carbon token)
- COAS: Carbon Offset Administration System (SA's national carbon registry)
- CBL: Carbon + Biodiversity Ledger (Xpansiv's trading platform)
- CT Clamp: Current Transformer clamp (non-invasive current sensor)
- DAO: Decentralized Autonomous Organization (community governance)
- DeFi: Decentralized Finance (blockchain-based financial services)
- D-REC: Distributed Renewable Energy Certificate
- ESP32: Low-cost microcontroller with WiFi/Bluetooth
- I-REC: International Renewable Energy Certificate
- JSE Ventures: JSE's carbon and REC trading platform (powered by Xpansiv)
- LoRaWAN: Long Range Wide Area Network (IoT communication protocol)
- mJ Token: Umbane's energy production token (1 mJ = 1 Wh)
- TradFi: Traditional Finance (centralized, regulated markets)
- T-REC: Tradable Renewable Energy Certificate (Government)
- VCU: Verified Carbon Unit (Verra's carbon credit)

Appendix B: Key Contacts

Umbane Project:

- GitHub: github.com/umbane/umbane
- Email: (to be added)

JSE Carbon Trading Desk:

- JSE Ventures: https://www.jse.co.za/carbon

Voluntary Bodies:

- zaREC: https://www.zarecs.co.za/
- RECSA: https://www.recsa.org.za

Regulatory Bodies:

- National Treasury (Carbon Tax): carbon.tax@treasury.gov.za
- DMRE (COAS): carbon.energy.gov.za
- FSCA: info@fsca.co.za
- ICASA: info@icasa.org.za

Carbon Consultants (South Africa):

- SouthSouthNorth: https://southsouthnorth.org/
- Climate Neutral Group SA: https://climateneutralgroup.com/en/za/
- Credible Carbon: https://www.crediblecarbon.com/

Appendix C: Further Reading

Carbon Markets:

- National Treasury (2025): "Carbon Markets in South Africa" (consultation paper)
- Verra: "VCS in South Africa" (https://verra.org/vcs-in-south-africa/)
- ENS Africa: "Launch of Voluntary Carbon Market in SA" (legal analysis)

Tokenization & DeFi:

- Toucan Protocol Docs: https://docs.toucan.earth/
- KlimaDAO Whitepaper: https://www.klimadao.finance/whitepaper
- Uniswap V3 Paper: https://uniswap.org/whitepaper-v3.pdf

Technical:

- Umbane GitHub: github.com/umbane/umbane
- OpenZeppelin Governor: https://docs.openzeppelin.com/contracts/governance
- Chainlink Oracles: https://docs.chain.link/

Regulations:

- White Paper on Renewable Energy Policy (November 2003)

Appendix D: Notes

1. Carbon Knowledge Hub (2025). "Offset use in South Africa's carbon tax". Bloomberg Finance L.P. Retrieved from https://www.carbonknowledgehub.com/factsheets/south-africa-carbon-tax-offset-use. Data as of June 3, 2025.

2. NFT Discussion: There is still ongoing discussion on relative merits of issuing NFT in a duel process or single process. In duel process, an mJ NFT is exchanged for aC EC20 (with process wrapped inside UMB EC20), in a single process, the entire procedure occurs automatically, with UMB deposited as an EC20 only after a carbon trade occurs.

Document Prepared For: Umbane Stakeholders

Date: March 20, 2026

Version: 1.0

Authors: Umbane Research Team

Contact: 082 4251454

Email: ubuntupunk@gmail.com

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